10.2 Calculus with Parametric Curves

Apply calculus to parametric curves to find slopes, lengths, and areas.

Tangent & Areas

A curve $x = f(t),\ y = g(t)$ is **differentiable** at t if f and g are differentiable at t.

If y is a differentiable function of x, then by Chain Rule, we have

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}.$$

If $\frac{dx}{dt} \neq 0 \Rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

If y is a twice-differentiable function of x and $dx/dt \neq 0$, then writing $y' = dy/dx$, we have

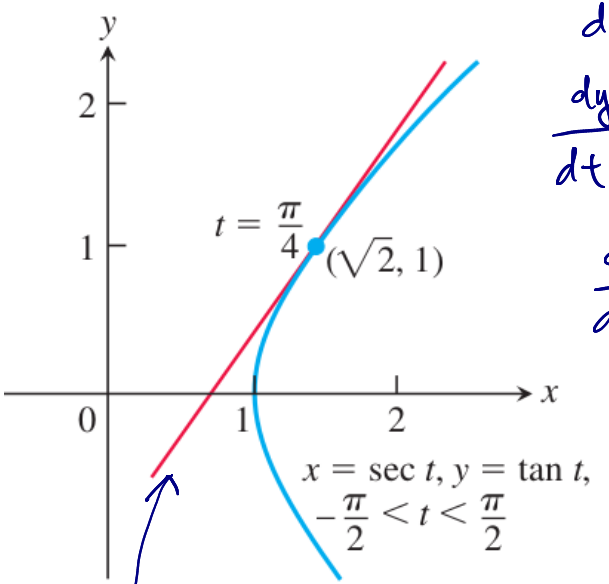
$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}.$$

$\left(y' = \frac{dy}{dx}\right)$

$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{dy'}{dt} \Big/ \frac{dx}{dt}$

$= \frac{dy'}{dx}$.

Example: Find the tangent to the curve $x = \sec t,\ y = \tan t,\ -\frac{\pi}{2} < t < \frac{\pi}{2}$, at the point $(\sqrt{2}, 1)$, where $t = \pi/4$.



$\frac{dx}{dt} = \sec t \tan t$

$\frac{dy}{dt} = \sec^2 t$.

$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

$= \frac{\sec t}{\tan t}$

$= \frac{1}{\sin t}$.

$x^2 = y^2 + 1$

$\Rightarrow$ diff wrt x

$\Rightarrow 2x = 2y \frac{dy}{dx}$

$\Rightarrow \frac{dy}{dx} = \frac{x}{y}$

@ $(\sqrt{2}, 1) \Rightarrow \left.\frac{dy}{dx}\right|_{\substack{x=\sqrt{2} \\ y=1}} = \sqrt{2}$

@ $\pi/4$

$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \frac{1}{\sin \pi/4} = \sqrt{2}$

$y - 1 = \sqrt{2}(x - \sqrt{2}) \Rightarrow \boxed{y = \sqrt{2}x - 1}$

Example: Find d^2y/dx^2 as a function of t if $x = t - t^2$ and $y = t - t^3$.

$$\frac{dy}{dt} = 1 - 3t^2 \qquad \frac{dx}{dt} = 1 - 2t$$

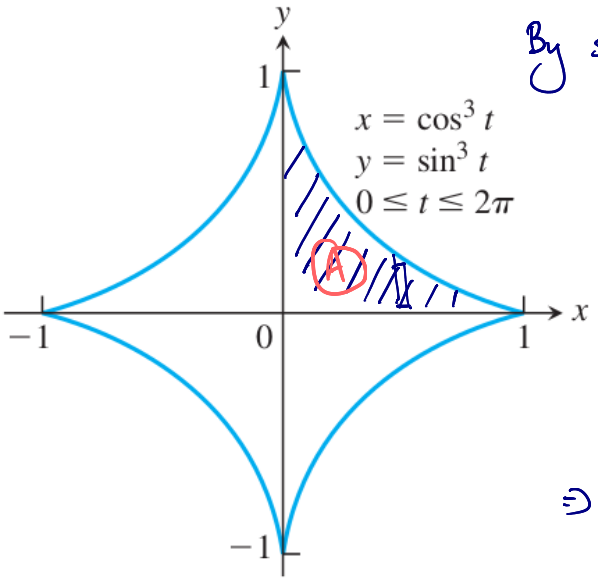
$$y' = \frac{dy}{dx} = \frac{1 - 3t^2}{1 - 2t}$$

$$\frac{dy'}{dt} = \frac{(-6t)(1-2t) - (-2)(1-3t^2)}{(1-2t)^2}$$

$$= \frac{2 - 6t + 6t^2}{(1-2t)^2}$$

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{2 - 6t + 6t^2}{(1-2t)^3}$$

Example: Find the area enclosed by the astroid $x = \cos^3 t,\ y = \sin^3 t,\ 0 \le t < 2\pi$.



By symmetry, the area of the astroid = 4A

$$A = \int_0^1 y\,dx$$

$$dx = 3\cos^2 t(-\sin t)\,dt$$

$x = 0 \Rightarrow t = \frac{\pi}{2}$

$x = 1 \Rightarrow t = 0$

$$\Rightarrow A = \int_{\frac{\pi}{2}}^{0} (\sin^3 t)(-3\cos^2 t \sin t)\,dt$$

$$= 3\int_0^{\frac{\pi}{2}} \sin^4 t \cos^2 t\,dt$$

$$= 3\int_0^{\frac{\pi}{2}} \left(\frac{1-\cos 2t}{2}\right)^2 \left(\frac{1+\cos 2t}{2}\right) dt$$

$$A = \frac{3}{8}\int_0^{\frac{\pi}{2}} (1 - \cos 2t - \cos^2 2t + \cos^3 2t)\,dt = \ldots = \frac{3}{8}\frac{\pi}{4}$$

∴ Area of astroid = 4A

$$= \frac{3\pi}{8}$$

Length of a Parametrically Defined Curve

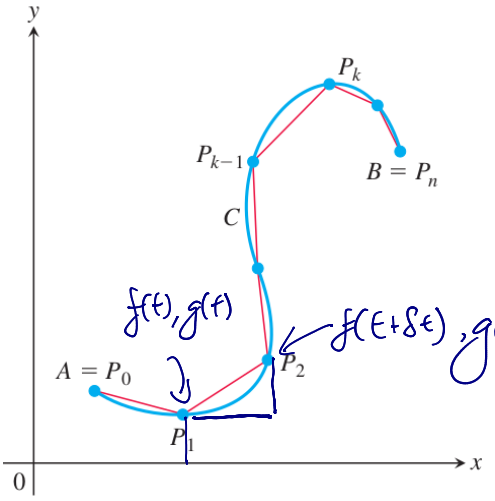
A curve C defined by $x = f(t)$ and $y = g(t)$, $a \le t \le b$ is a **smooth curve** if f and g are continuously differentiable with f' and g' not simultaneously zero.

Definition: Length of Smooth Curve

If C is a smooth curve defined by $x = f(t)$ and $y = g(t)$, $a \le t \le b$, and C is traversed exactly once at t increases from $t = a$ to $t = b$, then the **length of** C is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2}\, dt.$$

$$= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt .$$



$$\rightarrow \sqrt{\frac{(f(t+\delta t)-f(t))^2 + (g(t+\delta t)-g(t))^2}{(\delta t)^2}}\, \delta t$$

Example: Find the length of the circle of radius r defined parametrically by

$$x = r\cos t \quad \text{and} \quad y = r\sin t, \quad 0 \le t \le 2\pi.$$

$$L = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

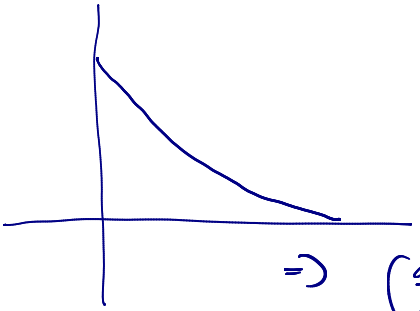
$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (-r\sin t)^2 + (r\cos t)^2$$

$$= r^2(\sin^2 t + \cos^2 t)$$

$$= r^2$$

$$\Rightarrow L = \int_0^{2\pi} r\, dt = \underline{\underline{2\pi r}}.$$

Example: Find the length of the astroid $x = \cos^3 t,\ y = \sin^3 t,\ 0 \le t < 2\pi$.



$$\frac{dx}{dt} = -3\sin t\cos^2 t$$

$$\frac{dy}{dt} = 3\sin^2 t\cos t$$

$$\Rightarrow \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\sin^2 t\cos^4 t + 9\sin^4 t\cos^2 t.$$

$$= 9\sin^2 t\cos^2 t(\cos^2 t + \sin^2 t)$$

$$= 9\sin^2 t\cos^2 t.$$

$$\Rightarrow \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = |3\sin t\cos t|$$

$$= 3\sin t\cos t \quad \left(\text{for } 0 \le t \le \frac{\pi}{2}\right)$$

← the first quadrant portion

length of the astroid in the first quadrant $= \int_0^{\frac{\pi}{2}} 3\sin t\cos t\,dt = \frac{3}{2}\int_0^{\frac{\pi}{2}} \sin 2t\,dt.$

$$= \frac{3}{4}\left[-\cos 2t\right]_0^{\pi/2} = \frac{3}{2}$$

$\therefore$ length of the astroid $= 4 \times \frac{3}{2} = \underline{\underline{6}}$